

Shubh Pathak

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Summary

AI Engineer at UBS with over 2 years building production ML pipelines and LLM systems for Investment Banking. Spearheaded a generative AI compliance-review platform processing 8,000+ documents monthly using Azure OpenAI, hybrid rule-LLM architectures, and async multi-threading. Collaborated with compliance and engineering stakeholders to deliver ML-driven automation across Investment Banking and Technology Services. Certified Azure AI Engineer (AI-102), Azure Data Scientist (DP-100), and TensorFlow Developer (Google).

Experience

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| UBS | Pune, India |
| AI Engineer | Nov 2025 – Present |
| <ul style="list-style-type: none">Led development of a generative AI compliance platform that eliminated 200+ hours of monthly manual review for a 6-person compliance team, processing 8,000+ Investment Banking documents monthly via two Python engines: a document engine (PDF, PPT, DOCX) with Azure OpenAI vision checks and a commentary engine for emails using grouped LLM prompt executionDevised a contextual routing strategy to solve an M x N scaling problem, cutting API calls by 67% and compute token cost by 45% by directing document chunks only to relevant compliance checks based on task complexityEngineered async multi-threading via ThreadPoolExecutor for parallel document parsing and model inference, slashing end-to-end pipeline latency by 70%Presented compliance KRI dashboards to Investment Bank leadership, translating real-time breach monitoring data into executive-ready insights; collaborated with compliance and risk teams to define check logic and KRI frameworks | |
| ML Engineer and Data Analyst | Aug 2024 – Oct 2025 |
| <ul style="list-style-type: none">Owned end-to-end development of a predictive ML model that automated approval and rejection decisions for 4,000+ IT infrastructure tickets annually, trained on historical virtual server utilization data covering CPU, memory, and disk metricsCompressed hardware approval cycles from 14 days to under 3 days by integrating telemetry from 85+ enterprise systems and deploying rule-based recommendation logic in PostgreSQLDecreased manual data-gathering effort by 40% by architecting the INSPIRE dashboard in Power BI, integrating datasets from 80+ enterprise systems into a unified infrastructure viewShipped the Solution Designer Studio in collaboration with business users, enabling cross-functional teams to accelerate infrastructure assessment cycles by 60% through custom reusable Power BI reports built with personalized visuals | |
| Data Analyst Intern | Jan 2024 – Jul 2024 |
| <ul style="list-style-type: none">Shrank mean-time-to-resolution from 2-3 days to under 6 hours by designing a Power BI failure-monitoring dashboard segmented by region, data source, and failure typeDrove migration of compliance reporting from Tableau to Power BI by automating Azure Data Factory ETL pipelines loading BigID scan outputs into MS SQL | |

Projects

- Ada – AutoML Agent** | *Python, LangGraph, FastAPI, React, OpenAI API* | github.com/shubh0614/ADA-MLFlow
- Architected a 9-agent LangGraph AutoML platform that automates the full ML workflow from data profiling to model deployment; shipped a FastAPI and React web UI, GitLab CI/CD pipeline integration, human-in-the-loop approval gate, and self-correcting LLM code execution with auto-retry on failure

Technical Skills

Machine Learning and AI: Python, TensorFlow, PyTorch, Azure OpenAI, LLMs, NLP, RAG, Prompt Engineering, LangGraph, Model Orchestration, Deep Learning

Data Engineering and Cloud: Azure Data Factory, PostgreSQL, MS SQL, pandas, PyMuPDF, openpyxl, Azure, AWS, FastAPI, Git, Power BI, Tableau, Stakeholder Management

Certifications

Microsoft Certified: Azure AI Engineer Associate (AI-102)	Microsoft, 2024
Microsoft Certified: Azure Data Scientist Associate (DP-100)	Microsoft, 2024
TensorFlow Developer Certificate	Google, 2024
Microsoft Certified: Fabric Analytics Engineer Associate (DP-600)	Microsoft, 2025

Education

Vellore Institute of Technology	Bhopal, India
Bachelor of Technology, Computer Science	Oct 2020 – Oct 2024
• GPA: 8.70 / 10.00	